Tristan Ozuch(-Meersseman)

Address : 182 Memorial Dr, Cambridge, MA 02142
Phone : +1 (617) 201-5392
Email : ozuch@mit.edu
Webpage : https://tristanozuch.github.io/

Main research interests

Geometric analysis, Einstein manifolds, Ricci Flows, 4-dimensional geometry and topology, applications to Physics.

Employment

- | **Assistant Professor**, *Rockwell International Career Development Professorship*, 2025–2028.
2023 | Massachusetts Institute of Technology, Cambridge, MA

2023 | **C.L.E. Moore Instructor**
2020 | Massachusetts Institute of Technology, Cambridge, MA

Education

2020 | **PhD in Mathematics at** ENS, Paris, France
2017 | Advisor : Olivier Biquard,
Subject : *Completion of the moduli space of Einstein 4-manifolds.*

2017 | **Graduated** from École Normale Supérieure (ENS), Paris, France

Grants and awards

2026–2027 **Global Seed Funds**, MIT-France – $30.000.
2025–2028 **Rockwell International Career Development Professorship**, MIT.
2024-2027 **NSF Grant: Geometric Analysis**, NSF DMS-2405328 – $200.000.
2024 **MIT Research Support Committee**, NEC corporation Fund – $90.000.
2021 **Young researcher fellowship at the Cluster of excellence**, Münster University.
2014-2019 **Travel grant FSMP and ENS**. 4 semester-long visits to American universities.
2017-2020 **PhD grant for graduate studies from ENS Ulm**.
2013-2017 **Grant for Undergraduate studies ENS Ulm**. 40 granted per year nationally.

Teaching

- **Assistant Professor**, MIT, Cambridge, MA, USA
  - Spring 2026: Course-head: **Differential equations** (18.03). 400 students from diverse departments.
  - Spring 2025: Course-head: **Seminar in Analysis** (18.104). 16 undergraduate and theses on optimal transport.
  - Fall 2024: Course-head: **Geometry of Manifolds I** (18.965), 30 undergraduate and graduate students.

2023
- Fall 2023: Course-head: **Linear Algebra** (18.06), 260 students from diverse departments.

2023 **C.L.E. Moore Instructor**, MIT, Cambridge, MA, USA
- Spring 2023: Course-head: **Differential Equations with theory** (18.032).
- Spring 2022: *Undergraduate class aimed at mathematics majors, 25 students*
- Spring 2021: *Redefined the syllabus and wrote my own lecture notes*
  In charge of a TA for recitations and a grader
- Fall 2022: Course-head: **Seminar in Analysis** (18.104).
- Fall 2021: *Undergraduate class on scientific writing and public presentations, 20 students*
  Chose the topic of **'Optimal Transport, Theory and Applications'**
  Supervised 8 theses in CS, 4 in pure maths , 3 in econ, 1 in scientific writing

2020
- Fall 2020: Teaching assistant: **Multivariable Calculus** (18.02).
  Undergraduate class for all science majors
  Led recitations for 60 students (300 total in the class)

2020 **Teaching assistant**, ENS, Paris, France

2017
- 2020: Course-head: **Mathematics for Humanities** (in French).
- 2019: *Designed a curriculum from scratch on topics in pure and applied mathematics*
- 2018: *Introductory lectures on scientific reasoning, geometry, number theory, machine learning, game theory*
- 2020: Coorganizer and jury of the **undergraduate theses in mathematics** at ENS.
- 2019: *Reached out to professors in the math department to create 20 thesis topics*
- 2018: *Put together a jury and evaluation grid, acted as like between thesis supervisors and students*

Publications and preprints

2026 *Blowdowns along Ricci flow in dimension four* (with D. Kim), arXiv:2611.00139 [math.DG].

2026 *Positive Einstein metrics on infinitely many four-manifolds*, arXiv:2610.08052 [math.DG].

2026 *Stability of gravitational instantons with a bounded Killing vector field*, arXiv:2608.31008 [math.DG].

2026 *Desingularizations of conformally Kähler, Einstein orbifolds* (with C. LeBrun), arXiv:2601.19215 [math.DG].

2025 *Regularity of Einstein 5-manifolds via 4-dimensional gap theorems* (with Y. Huang), arXiv:2512.21317 [math.DG].

2025 *Linear stability and instability of Kähler Ricci solitons* (with K. Naff), accepted, to appear in **Communications in Analysis and Geometry**, arXiv:2511.15885 [math.DG].

2025 *Ricci flow on ALF manifolds* (with D. Kim), arXiv:2510.21997 [math.DG].

2025 *Linear stability of the blowdown Ricci shrinker in 4D* (with K. Naff), arXiv:2509.05470 [math.DG].

2024 *Orbifold singularity formation along ancient and immortal Ricci flows* (with A. Deruelle), arXiv:2410.16075 [math.DG].

2024 *Ancient and expanding spin ALE Ricci flows* (with I. M. Lopez), **J. Funct. Anal. 289** (2025), no. 9, Article 111062.

2023 *Instability of conformally Kähler, Einstein metrics* (with O. Biquard), **J. Differential Geom. 133** (2026), no. 3, 335–346.

2022 *The spinorial energy for asymptotically Euclidean Ricci flow* (with J. Baldauf), **Adv. Nonlinear Stud.** (Special Issue: Geometric PDEs and Applications) **23** (2023), Article 20220045.

2022 *Families of degenerating Poincaré–Einstein metrics on* $\mathbf{R}^4$ (with C. A. Alvarado and D. A. Santiago), **Ann. Global Anal. Geom. 65** (2024), Article 5.

2022 *Spinors and mass on weighted manifolds* (with J. Baldauf), **Commun. Math. Phys. 394** (2022), 1153–1172.

2021 *Integrability of Einstein deformations and desingularizations*, **Commun. Pure Appl. Math. 77** (2024), 177–220.

2021 *Dynamical (in)stability of Ricci-flat ALE metrics along Ricci flow* (with A. Deruelle), **Calc. Var. Partial Differential Equations 62** (2023), Article 84.

2021 *Depth separation beyond radial functions* (with L. Venturi, S. Jelassi and J. Bruna), **J. Mach. Learn. Res. 23** (2022), no. 122, 1–56.

2020 *Higher order obstructions to the desingularization of Einstein metrics*, **Camb. J. Math. 9** (2021), no. 4, 901–976.

2020 *Completion of the moduli space of Einstein* 4*-manifolds*, Ph.D. thesis, **École normale supérieure (Paris)**, 2020.

2020 *A ojasiewicz inequality for ALE metrics* (with A. Deruelle), **Ann. Sc. Norm. Super. Pisa Cl. Sci. 26** (2025), no. 4, 1965–2042.

2019 *Noncollapsed degeneration of Einstein* 4*-manifolds II*, **Geom. Topol. 26** (2022), 1529–1634.

2019 *Noncollapsed degeneration of Einstein* 4*-manifolds I*, **Geom. Topol. 26** (2022), 1483–1528.

2019 *Perelman's functionals on cones and construction of type III Ricci flows coming out of cones*, **J. Geom. Anal. 30** (2020), 1–53.

2019 *How large isotopy is needed to connect homotopic diffeomorphisms (of* $\mathbf{T}^2$*)* (with D. Burago and J. Lu), **J. Topol. Anal. 12** (2020), no. 4, 1213–1222.

Editorial service

Editor at *Bulletin of the London Mathematical Society* and *Journal of the London Mathematical Society*.

Referee service

Journal of Differential Geometry, GAFA, Crelle's journal, Annales de l'École Normale Supérieure, Journal of Geometric analysis, Revista Mathematica Iberoamericana, Mathematische Zeitschrift, Mathematische Annalen, American Journal of math, Calc. Var. PDE, Pacific J. Math., SIGMA, Proceedings AMS, Transactions AMS, Annales de l'institut Fourier, Mathematical Reviews, zbMATH Open.

List of students mentored

Graduate students

- Joseph Bonavia, MIT (co-advised with T. Cohen, A. Goriely, D. Parks)

Jan. 2025 .

- Yiqi Huang, MIT (co-advised with T. Colding)

Sept. 2024 *Regularity of Einstein* 5*-manifolds* (one joint preprint).

- Cosmin Manea, MIT (co-advised with T. Colding)

Sept. 2024 .

- Benjy Firester, MIT (co-advised with T. Collins and T. Colding)

Sept. 2023 .

- Dain Kim, MIT (co-advised with B. Minicozzi)

Sept. 2023 *Ricci flow on ALF metrics* (two joint preprints).

Sept. 2023 Julius Baldauf, MIT (co-advised with B. Minicozzi)

Sept. 2020 *Ricci flows and spin geometry* (two published paper in common, one on his own).

Undergraduate students

Spring 2025 Junhwi Bae, MIT (UROP)

Fall 2025	Junhwi Bae, MIT (UROP)
Summer 2025	Katie Hall, William Ewald, Junhwi Bae, MIT (UROP)
Spring 2025	Katie Hall, William Ewald, Isaac Lopez, Junhwi Bae, MIT (UROP)
Fall 2024	Katie Hall, William Ewald, Isaac Lopez, Junhwi Bae, MIT (UROP)
Summer 2024	Katie Hall, William Ewald, MIT (UROP)
Spring 2024	Isaac Lopez, MIT (UROP)
Fall 2023	Isaac Lopez, MIT (UROP)
Summer 2023	Daniel Santiago & Isaac Lopez, MIT (SPUR+)
Summer 2023	Enrique Rivera, MIT (UROP)
Summer 2021	Carlos Alvarado & Daniel Santiago, MIT (UROP)
Spring 2021	Carlos Alvarado & Daniel Santiago, MIT (UROP).
Fall 2022	Zachary Hunsucker, MIT (UROP) *Schrödinger bridges along Ricci flow and quantum optimal transport*.
Spring 2019	Raphaël Barboni, Haohao Liu & Martin Malvy, ENS (bachelor thesis) *Level set methods for mean curvature flow*.

Committee

Graduate theses committees

Julius Baldauf (2024), Tang-Kai Lee (2025), Xinrui Zhao (2025).

Research visits

August 2021 July 2021	**Young Research Fellow at University of Münster** *Invited by* : Hans-Joachim Hein Studied toric Einstein metrics.
December 2019 September 2019	**Graduate Visiting Student at Courant institute** *Advisor* : Bruce Kleiner Desingularization of Einstein manifolds and Ricci flows.
July 2017 February 2017	**Research Internship at Northwestern University** *Advisor* : Aaron Naber Study of the degeneration of Einstein 4-manifolds.
February 2017 September 2016	**Research internship at ENS** *Advisor* : Olivier Biquard Study of the desingularization of Einstein orbifolds and obstructions
July 2016 February 2016	**Research Internship at UC Berkeley/MSRI** *Advisor* : Richard Bamler
July 2015 February 2015	**Research Internship at Penn. State University** *Advisor* : Dmitri Burago

Seminars and conferences

As an invited speaker :

July 2027	**Differentialgeometrie im Grossen**, Oberwolfach.
Spring 2027	**Differential Geometry Seminar**, UC Berkeley.
Spring 2027	**Geometry/Topology Seminar**, Duke University.
December 2026	**Differential Geometry and Mathematical Physics**, CMS Winter Meeting, short talk.
December 2026	**Einstein 4-manifolds and their degenerations**, CMS Winter Meeting, minicourse.
November 2026	**PDE and Differential Geometry Seminar**, University of Connecticut

November 2026 **Colloquium**, Stony Brook University.

October 2026 **Geometria em Lisboa**, online.

October 2026 **Séminaire d'Analyse et Géométrie**, Jussieu, Paris.

June 2026 **Geometrie**, Oberwolfach

May 2026 **International Conference on Complex & Differential Geometry**, ICTP

May 2026 **Living on the edge of the moduli space**, Aarhus University

April 2026 **Differential Geometry & Geometric Analysis Seminar**, Princeton University.

March 2026 **CMSA Differential Geometry and Physics Seminar**, Harvard University

January 2026 **Einstein 4-Manifolds and Gravitational Instantons**, Simons center

December 2025 **Geometry seminar**, IST Lisbon.

June 2025 **Master Class series**, Université de Nantes

May 2025 **Geometric Analysis Seminar**, University of Chicago.

April 2025 **Workshop on Canonical Metrics in Differential Geometry**, University of Wisconsin Madison.

April 2025 **Geometry/Topology Seminar**, Stony Brook University.

January 2025 **Théorie spectrale et géométrie**, Institut Fourier, Grenoble

January 2025 **Séminaire d'Analyse et Géométrie**, Jussieu, Paris

January 2025 **Journées parisiennes d'analyse géométrique**, Université Gustave Eiffel

November 2024 **Curvature and Geometric Analysis in Rome**, Sapienza Univ. di Roma.

October 2024 **Geometry seminar**, Michigan State University

May 2024 **Caltech Geometry & Topology Seminar**, Caltech.

May 2024 **Workshop on Analysis of Geometric Singularities**, CRM in Montréal.

April 2024 **Math department Seminar**, Fordham University.

April 2024 **Geometry/Topology Seminar**, Stony Brook University.

March 2024 **Joint Princeton-Rutgers Geometric PDEs Seminar**.

March 2024 **Geometry and Topology Seminar**, University of Wisconsin Madison.

February 2024 **Geometry Seminar**, Stanford University.

February 2024 **Lehigh Geometry Seminar**, Lehigh University.

February 2024 **Analysis, Geometry and Topology of PSC Metrics**, Oberwolfach.

February 2024 **Differential Geometry Seminar**, UC Berkeley.

December 2023 **Séminaire d'analyse**, Toulouse, France.

November 2023 **Recent advances in geometric analysis**, CIRM, Marseilles.

December 2023 **Geometric PDE**, Canadian Math society, Winter meeting, Montréal.

October 2023 **Analysis seminar**, Columbia university.

September 2023 **BU Geometry and Physics Seminar**, Boston.

July 2023 **Analysis, Geometry and Topology of PSC Metrics**, Oberwolfach.

August 2023 **Analytic Methods in Complex Geometry**, University of Münster.

August 2023 **Workshop On curvature and global shape**, University of Münster.

July 2023 **Differentialgeometrie im Grossen**, Oberwolfach.

July 2023 **Conf. on Einstein spaces and special geometry**, Inst. Mittag-Leffler.

April 2023 **Geometric analysis Seminar** , Rutgers University

April 2023 **PDE and Differential Geometry Seminar** , University of Connecticut

March 2023 **Conference on Geometric Analysis**, Regensburg

March 2023	**Ricci flow and related topics** , Warwick
January 2023	**Colloquium of mathematics**, University of Notre Dame
January 2023	**Colloquium of mathematics**, UC Davis
January 2023	**Pure math seminar**, MIT
December 2022	**Colloquium of mathematics**, Lehigh University
November 2022	**Differential Geometry seminar**, UC Berkeley.
November 2022	**Geometry/Topology seminar**, Stony Brook University.
June 2022	**Canadian Mathematical Society Meeting**, Newfoundland.
May 2022	**CMSA workshop**, Harvard University.
April 2022	**Metric measure spaces and convergence**, University of Mexico.
April 2022	**Geom. & Top. seminar**, Univ. of Science and Technology of China.
April 2022	**KIT Geometric Analysis Seminar**, KIT.
February 2022	**Cornell Analysis and Geometric analysis seminar**, Cornell University.
February 2022	**BOWL Seminar**, Brussels, Oxford, Warwick, London.
October 2021	**Geometric analysis seminar**, UCL.
October 2021	**Séminaire d'analyse**, Université de Toulouse.
October 2021	**Geometry Topology Dynamical Systems Seminar**, UT Dallas.
October 2021	**Geometric analysis seminar**, MIT.
September 2021	**Geometric analysis seminar**, Yale University.
August 2021	**Workshop on Curvature and Global Shape**, University of Münster.
July 2021	**International Conference**, Princeton University - Shanghai Jiaotong.
July 2021	**Curvature constraints and spaces of metrics**, Institut Fourier.
May 2021	**Simons Collaboration on Special Holonomy.** Simons center.
March 2021	**Geometric Analysis Seminar**, University of Chicago.
March 2021	**Differential Geometry & Geometric Analysis Seminar**, Princeton Univ.
November 2020	**Seminar: Cheeger-Colding theory, Ricci flow, Einstein metrics**, UCSD.
November 2020	**Stanford's Geometry seminar**, Stanford University.
November 2020	**BOWL Seminar**, Brussels, Oxford, Warwick, London.
June 2020	**Oberseminar of Differential Geometry**, Münster.
February 2020	**Séminaire Darboux**, Montpellier.
December 2019	**Differential Geometry Seminar**, UC Berkeley.
November 2019	**Geometric Analysis and Topology Seminar**, Courant institute, NYU.
October 2019	**Geometry/Topology Seminar**, Stony Brook University.
May 2019	**Convergence and Low Regularity in General Relativity**, Simons Center.
March 2019	**Geometry seminar**, Bruxelles.
February 2019	**Geometry seminar**, Nantes.
March 2018	**Masters-PhD meeting**, Jussieu.
February 2018	**Geometry seminar of IMJ**, Paris Diderot.
January 2018	**Graduate students seminar**, ENS.
April 2016	**Graduate student seminar**, MSRI, Berkeley.

As a participant :

September 2022	**Special Holonomy: Progress and Open Problems 2022** Simons Center, Stony Brook University

September 2022	**Sixth Annual Meeting: collaboration on special holonomy** Simons Foundation, NYC
June 2022	**Simons collaboration meeting** University of Freiburg
May 2021	**Atelier sur la Géométrie différentielle et l'analyse globale** UQAM
October 2019	**Recent advances in nonlinear problems Symposium** Graduate Center, CUNY
May 2019	**Master class in differential geometry : the structure of limit spaces** Institut Henri Poincaré, Paris
December 2018	**Geometric analysis at IHP** Institut Henri Poincaré, Paris
27 July 2018 23 July 2018	**McGill University Geometric Analysis Workshop 2018** McGill University, Montréal
1 June, 2018 28 May, 2018	**Geometric Analysis** ICMS, Edimburgh
16 February 2018	**Géométrie : échanges et perspectives** Institut Henri Poincaré, Paris
9 December 2017 6 December 2017	**Riemannian Geometry Past, Present and Future: an homage to Marcel Berger**, IHES, Bures-sur-Yvette
13 October 2017 9 October 2017	**Conference - Geometric Analysis at Roscoff**, Centre Henri Lebesgue, Roscoff
21 July 2017 10 July 2017	**Summer school in Geometric Analysis**, The Fields institute, Toronto "Ricci flow and intrinsic flat convergence" research team.
July 2016 February 2016	**Differential geometry semester at MSRI**, MSRI, Berkeley

Other Skills

Languages
French : Native speaker.
English : Fluent.
Italian : Good command.

Activities
Swimming, running, whittling.